

Q1. Create an abstract Payment class and implement Credit Card and UPI payment classes.

Code:

```
// Abstract class
abstract class Payment
{
    abstract void pay(double amount);
}

// Credit Card payment
class CreditCard extends Payment
{
    void pay(double amount)
    {
        System.out.println("Payment Mode: Credit Card");
        System.out.println("Amount Paid: Rs. " + amount);
    }
}

// UPI payment
class UPI extends Payment
{
    void pay(double amount)
    {
        System.out.println("Payment Mode: UPI");
        System.out.println("Amount Paid: Rs. " + amount);
    }
}

public class PaymentDemo
{
    public static void main(String[] args)
```

```

{
    // Parent class reference pointing to child class objects
    Payment p1 = new CreditCard();
    Payment p2 = new UPI();

    p1.pay(2500);
    System.out.println();

    p2.pay(1500);
}
}

```

Q2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

Code:

```

// Abstract class
abstract class FoodOrder
{
    double price;

    FoodOrder(double price)
    {
        this.price = price;
    }

    abstract void calculateBill();
}

```

```
}

// Dine-In order
class DineInOrder extends FoodOrder
{
    DineInOrder(double price)
    {
        super(price);
    }

    void calculateBill()
    {
        double serviceCharge = price * 0.05;
        double total = price + serviceCharge;

        System.out.println("Order Type: Dine-In");
        System.out.println("Food Price: Rs. " + price);
        System.out.println("Service Charge: Rs. " +
serviceCharge);
        System.out.println("Total Bill: Rs. " + total);
    }
}

// Take-Away order
class TakeAwayOrder extends FoodOrder
{
    TakeAwayOrder(double price)
    {
        super(price);
    }

    void calculateBill()
    {
        double packagingCharge = 30;
        double total = price + packagingCharge;

        System.out.println("Order Type: Take-Away");
```

```
        System.out.println("Food Price: Rs. " + price);
        System.out.println("Packaging Charge: Rs. " +
packagingCharge);
        System.out.println("Total Bill: Rs. " + total);
    }
}

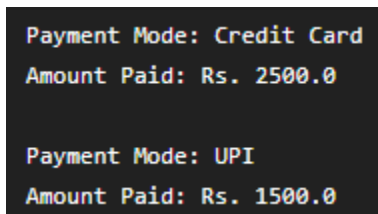
public class FoodOrderDemo
{
    public static void main(String[] args)
    {
        // Polymorphism using parent class reference
        FoodOrder order1 = new DineInOrder(500);
        FoodOrder order2 = new TakeAwayOrder(500);

        order1.calculateBill();

        System.out.println();

        order2.calculateBill();
    }
}
```

Q1 Screenshot



Payment Mode: Credit Card
Amount Paid: Rs. 2500.0

Payment Mode: UPI
Amount Paid: Rs. 1500.0

Q2 Screenshot

Order Type: Dine-In
Food Price: Rs. 500.0
Service Charge: Rs. 25.0
Total Bill: Rs. 525.0

Order Type: Take-Away
Food Price: Rs. 500.0
Packaging Charge: Rs. 30.0
Total Bill: Rs. 530.0